

Design Documentation: Hardware Hustle

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Project: Hardware Hustle - Robotic Simulation Essentials

1. Executive Summary

This document serves as the primary design record for our team's submission to the Hardware Hustle challenge. It details the engineering decisions, custom fabrications, and circuitry designs implemented to solve the arena challenges. Our approach focuses on reliability, structural optimization, and manufacturing efficiency. Key innovations include a topology-optimized aerodynamic canopy for the ALFR, a rigid yet lightweight custom wooden base for the SARM, and a simplified differential drive system for enhanced stability during object manipulation.

2. Circuitry Design of the Arena (Reference S.No 1)

The arena circuitry is the central nervous system of the challenge, managing the fluid delivery, visual feedback, and final weight verification. We have designed a centralized control system using the **ESP32 Development Board** provided in the kit, which offers sufficient GPIOs for the required sensors and actuators.

2.1 System Architecture

The circuit is divided into three logical "Gates" as defined by the rulebook, all controlled by the central ESP32 unit.

- **Central Controller:** ESP32-WROOM-32
- **Power Management:** 12V Source -> LM2596 Buck Converter -> 5V System Bus (Powering ESP32, Sensors, and Servos).

2.2 Operational Logic & Connections

Gate 1: Fluid Delivery System

- **Sensor:** IR Obstacle Avoidance Sensor (Digital Input). Detects the arrival of the ALFR.
- **Actuation:** L298N Motor Driver (Channel A).
- **Logic:** Upon IR trigger (LOW signal), the ESP32 activates the Peristaltic Pump motor via the L298N driver for a calibrated duration (calculated based on flow rate) to dispense exactly **125 ml** of fluid.
- **Status:** System waits for completion before resetting for the next cycle.

Gate 2: LAM LED Activation

- **Sensor:** IR Sensor Array (Analog/Digital).
- **Actuation:** 5V Relay Module.
- **Logic:** As the ALFR breaks the beam at Gate 2, the relay triggers the "LAM" LED array. The use of a relay isolates the high-brightness LED load from the microcontroller's logic pins.

Gate 3: Parking & LCD Display

- **Sensor:** 5Kg Load Cell with HX711 Amplifier Module.
- **Display:** 1.8 Inch TFT LCD Module (SPI Interface).
- **Logic:**
 1. ALFR parks on the weighing platform.
 2. Load cell reads the total weight (Robot + Container + Fluid).
 3. If weight is within tolerance: LCD displays "**Team Name**" and "**Simulation Completed!**".
 4. If weight mismatch: LCD displays "**Weight Mismatch**".

3. Single Arm Robot (SARM) Design (Reference S.No 2)

The SARM is designed for manual operation with a primary focus on stability and lifting capacity. While the rulebook suggests an Omni-wheeled platform, our analysis indicated that traction and structural rigidity were higher priorities for the obstacle removal task.



3.1 Customization: Wooden Chassis

Instead of a standard 3D-printed chassis, we engineered a **customized base made of wood**.

- **Material Selection:** Wood was chosen for its excellent strength-to-weight ratio and natural vibration dampening properties.
- **Advantages:**
 - **Rigidity:** The wooden base provides a stiffer platform than PLA, reducing flex when the arm lifts heavy obstacles.
 - **Manufacturability:** It allowed for rapid prototyping and easy modification of mounting points without long 3D printing queues.
 - **Weight:** It remains lightweight, ensuring the motors are not over-torqued.



3.2 Customization: Wheel Configuration

We opted to use **Standard Robot Wheels** (Differential Drive) instead of Mecanum wheels.

- **Design Rationale:** Mecanum wheels require complex vectoring and can slip on smooth arena surfaces, potentially compromising the precision needed to grab an obstacle.
- **Performance:** The standard wheels provide superior traction and predictable kinematics. This ensures that when the operator commands the robot to stop and lift, the base remains planted, giving the arm a stable foundation.

3.3 Customization: Gripper Mechanism

We developed a **customized gripper** specifically optimized for the challenge's obstacles.

- **Mechanism:** The gripper utilizes high-torque servos (MG996R) with a linkage design that converts rotary motion into linear clamping force.

- **Benefit:** The jaw geometry is contoured to match the obstacles, allowing the SARM to lift them "with ease" and minimal operator effort.

4. Advanced Line Follower Robot (ALFR) Design (Reference S.No 3)

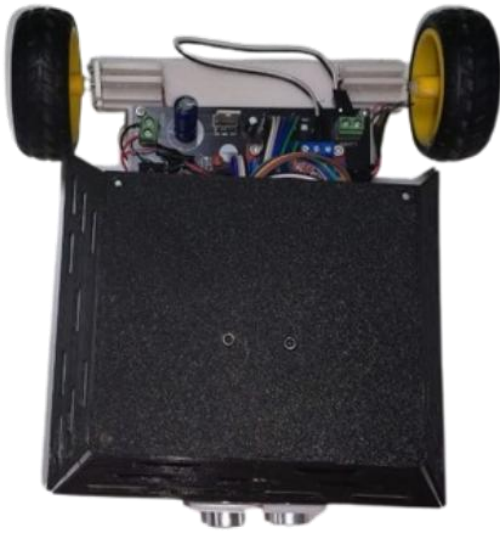
The ALFR is our autonomous unit. The core chassis is 3D printed, but we have significantly upgraded the upper housing and the control algorithm to ensure smooth, reliable performance.



4.1 Customization: Topology Optimized Canopy

We designed and fabricated a custom canopy for the ALFR.

- **Topology Optimization:** Using computer-aided engineering (CAE) tools, we optimized the material distribution of the canopy. This process removed material from low-stress areas while reinforcing high-stress load paths.
- **Aerodynamics:** The shape is streamlined to reduce drag, although at low speeds this primarily serves an aesthetic function, giving the robot a sleek, modern "Hardware Hustle" identity.
- **Function:**
 - **Protection:** It fully encloses the PCB and wiring, protecting them from loose wires or accidental impacts.
 - **Lightweight:** The topology optimization ensures the cover adds negligible weight to the robot, preserving battery life and speed.



4.2 Control Algorithm: PID Tuning

To ensure the ALFR moves smoothly and handles curves with ease, we implemented a **PID (Proportional-Integral-Derivative) control algorithm** rather than simple "bang-bang" (left/right) logic.

- **Proportional (P):** Handles immediate error correction. If the robot drifts slightly off the line, the P-term applies a proportional correction to steer it back.
- **Integral (I):** Accounts for accumulated error, helping the robot maintain a straight path even if there are minor mechanical imbalances in the motors.
- **Derivative (D):** Predicts future error by analyzing the rate of change. This effectively "dampens" the movement, preventing the robot from overshooting the line and oscillating wildly.
- **Result:** The PID loop calculates precise motor speeds thousands of times per second. This results in fluid, organic movement where the robot glides along the track—even through tight turns—without the jerky, robotic motion typical of basic line followers.

5. Peristaltic Pump Design (Reference S.No 7)

For the fluid delivery challenge, we designed and developed a custom peristaltic pump head driven by a DC motor.

5.1 Working Principle

The pump operates on the principle of **positive displacement**.

1. A flexible silicone tube is fitted inside a circular casing.
2. A central rotor with three rollers compresses the tube.
3. As the rotor turns, the rollers pinch the tube closed and force the fluid forward (peristalsis).
4. The restitution (expansion) of the tube behind the roller creates a vacuum that draws more fluid in.

5.2 Design Specifications

- **Tube:** Medical-grade silicone (high flexibility, chemical resistance).
- **Rollers:** 3D printed rollers with bearings to reduce friction on the tube.
- **Flow Calibration:** We conducted timed tests to determine the flow rate (ml/second). The ESP32 logic uses this calibration to run the pump for the exact duration needed to dispense 125 ml.
- **Benefit:** This design ensures the fluid never touches the pump mechanism, preventing contamination and ensuring a precise, non-siphoning stop when the motor cuts power.

6. Conclusion & Identity (Reference S.No 8)

This design document reflects our team's identity: **Pragmatic Innovation**. We did not simply follow the default instructions; we analyzed the problem and engineered better solutions. From the **topology-optimized canopy** of the ALFR that blends aesthetics with engineering rigor, to the **PID-tuned control system** that ensures silky-smooth navigation, every component has been deliberately chosen to ensure success in the Hardware Hustle arena.